

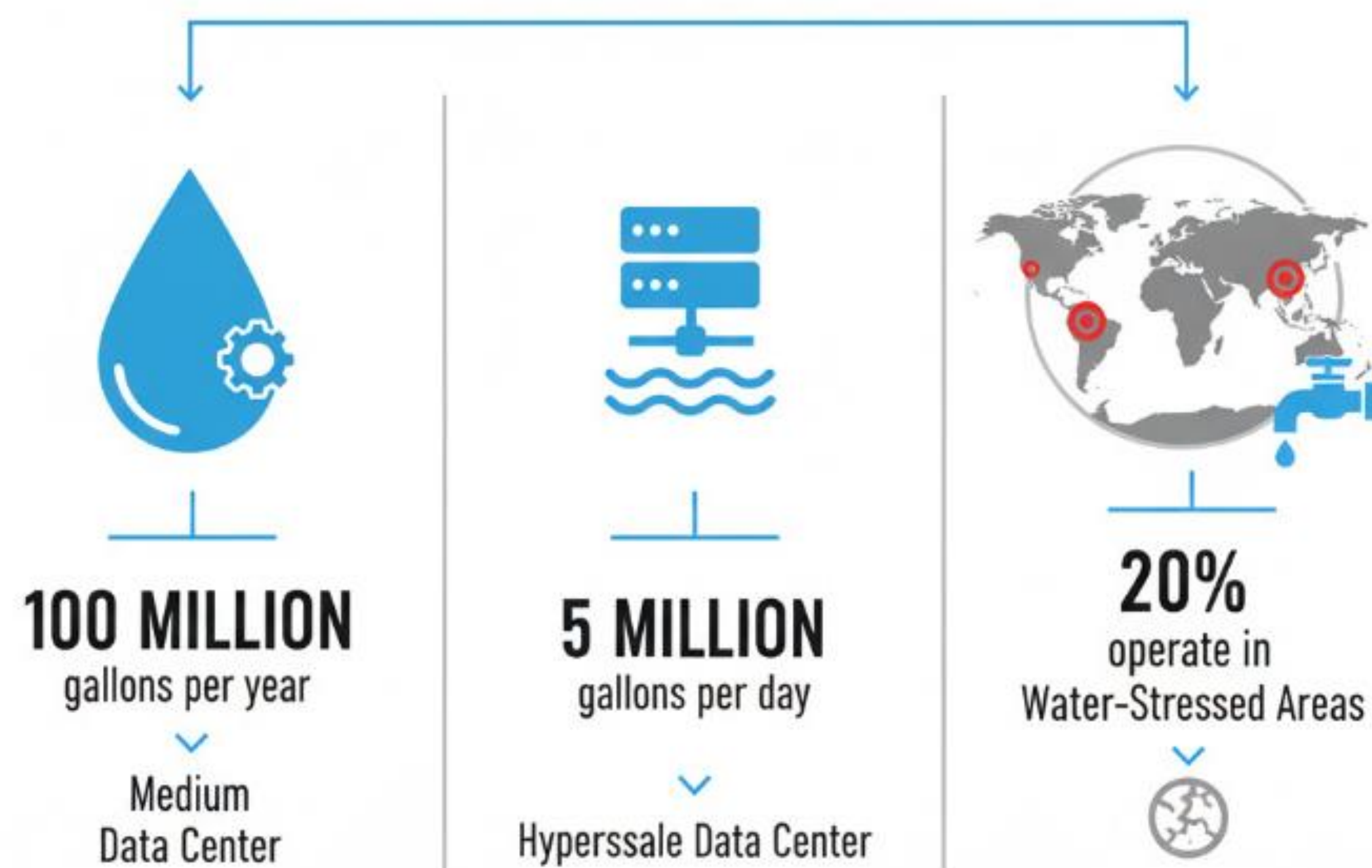


BUBBLE-DRIVEN AI DATA CENTER COOLING: *TWO-PHASE IMMERSION COOLING (2-PIC) SOLUTION*

OVERVIEW

- AI and high-performance computing are generating more heat than traditional data center cooling can handle
- As chip power continues to increase, conventional cooling methods are becoming less efficient
- Two-phase immersion cooling (2-PIC) uses boiling liquid to absorb and remove heat much more effectively
- Approach allows for higher computing density and improved energy efficiency
- Potential to reduce cooling costs and environmental impact in data centers
- Our team is studying how effective, scalable, and sustainable these systems are for future use

DATA CENTER WATER CONSUMPTION FACTS



Why Research Two-Phase Cooling?

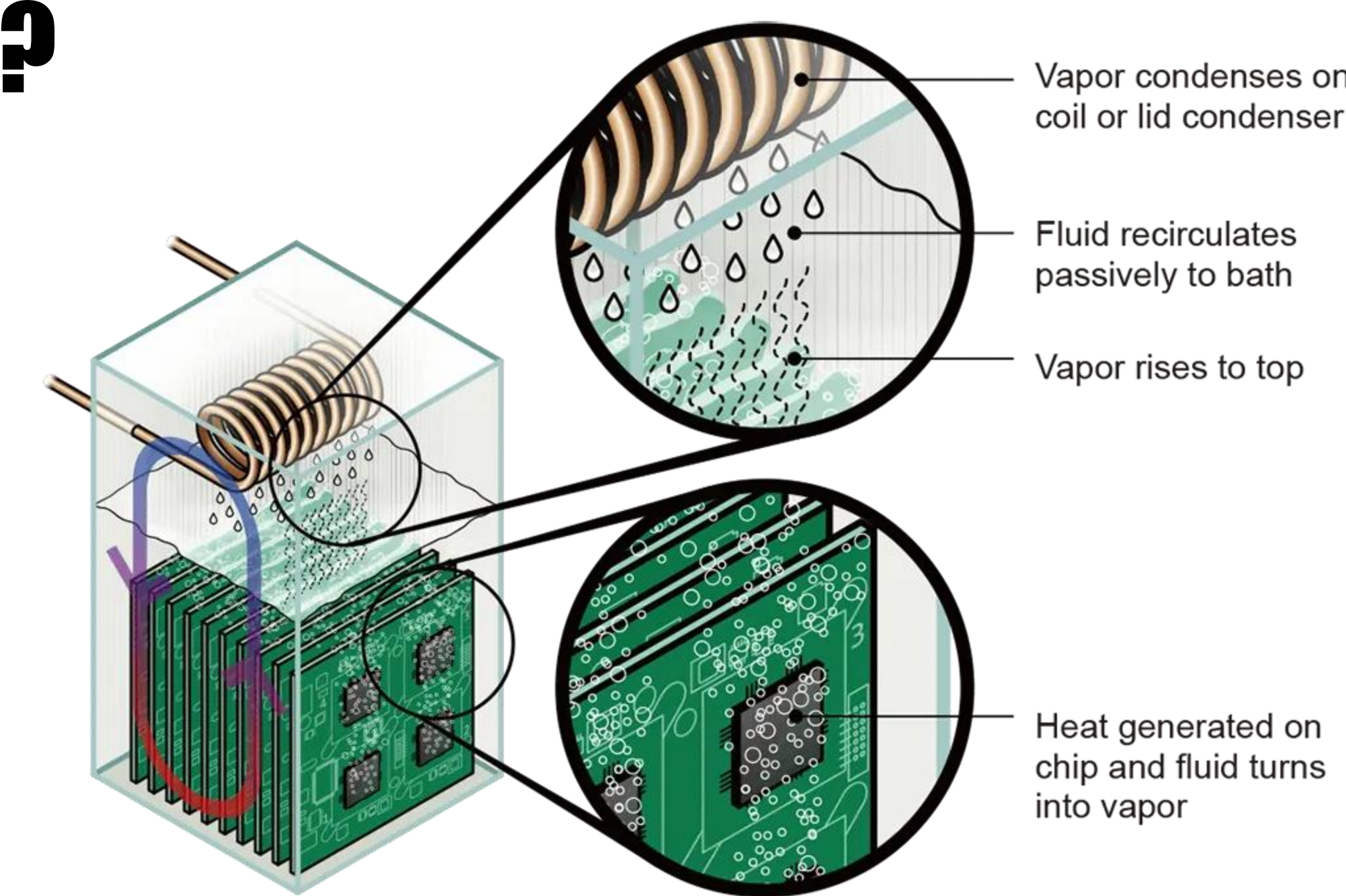
The A.I. industry was \$189 billion as of 2023, and according to UNCTAD it will increase to a \$4.8 Trillion dollar industry by 2033. This growth is matched with the emergence of new data centers with Meta, Amazon, and Microsoft investing 22\$ billion into Indiana Data Centers.

Data centers effect their local communities. The 1\$ billion data center in La Porte Is set to create 200 new jobs by 2032 and the taxes on the center are to go to the local schools. However, historically larger data centers consume up to 5 million gallons of water daily and can consume between 20-100 MW of electricity continuously. This water consumption, and a good portion of electricity consumption comes from inefficient water cooling.

As of 2023 4.4% of the united States electricity consumption came from data centers. It is projected to reach between 6.7% - 12% by 2028.

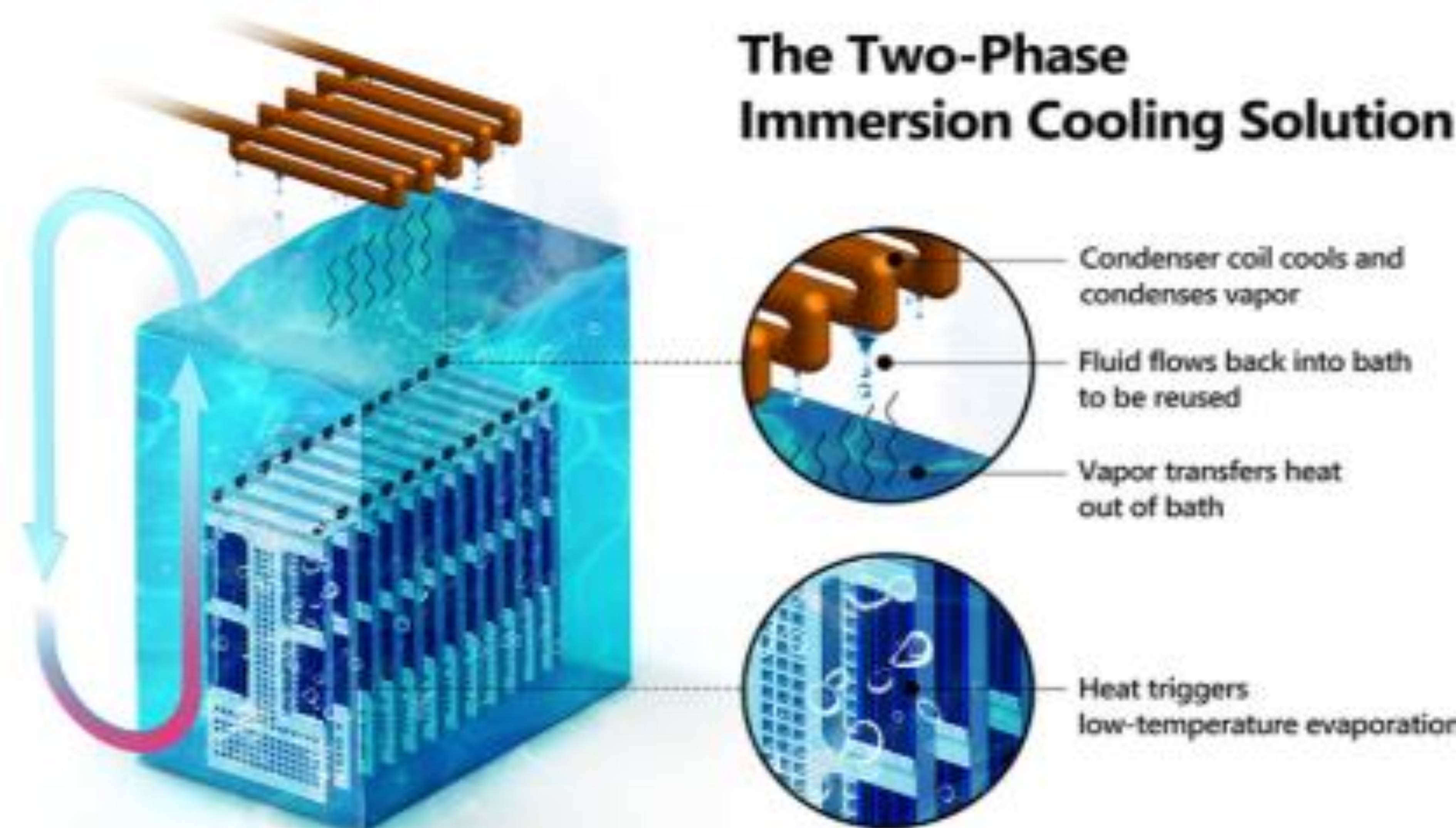
What is Two-Phase Cooling?

Two-Phase Immersion Cooling (2-PIC) is an advanced, highly efficient data center cooling technology where servers are submerged in a dielectric (non-conductive) liquid that boils to remove heat via the very high latent-heat-of-vaporization. This phase change (liquid-to-gas) provides superior thermal management, allowing 90% higher energy efficiency, resulting massive space savings (30x higher heat removal than air cooling systems) and higher computing density.



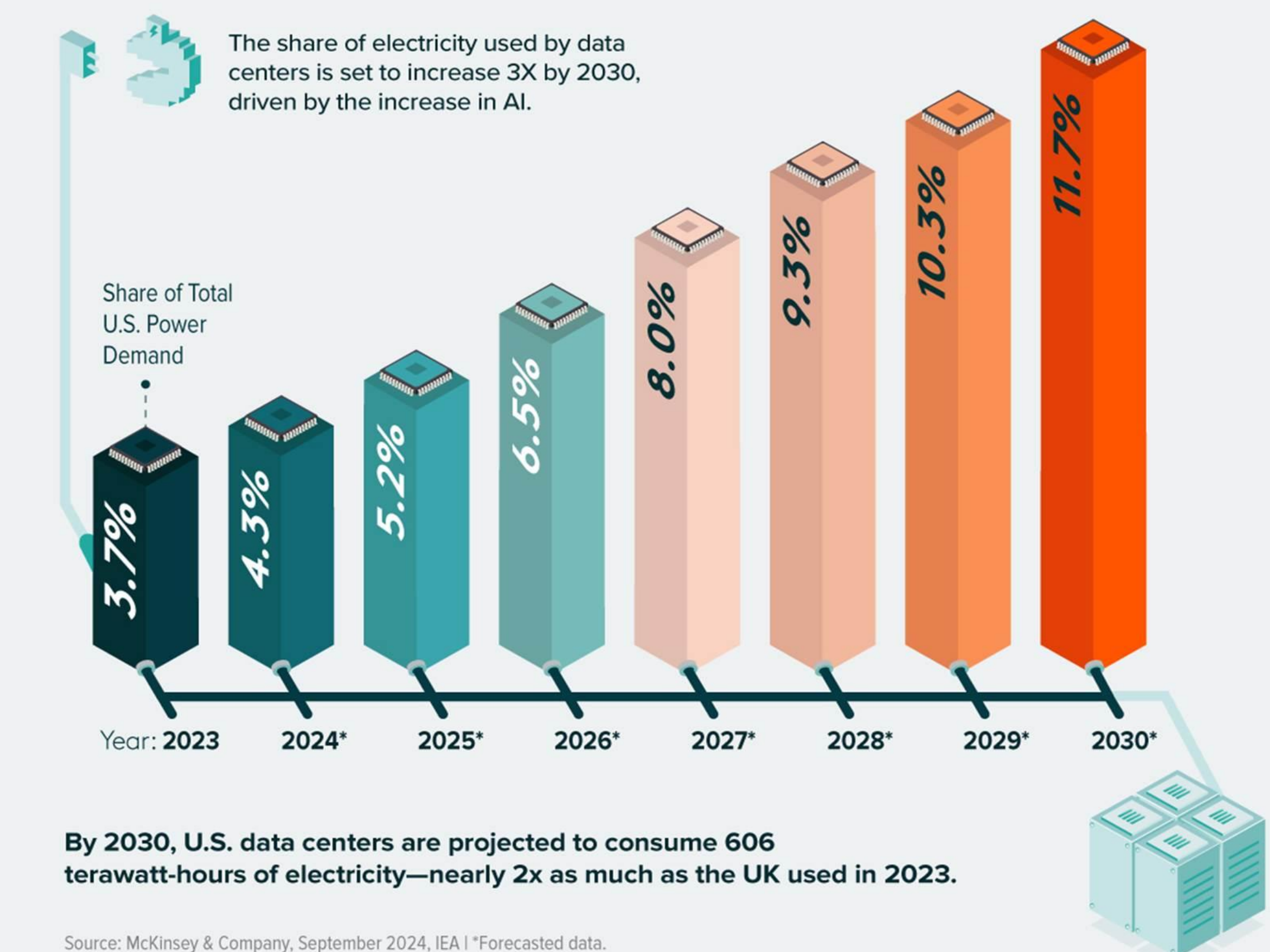
The Passive 2-Phase Immersion Cooling Cycle

The Two-Phase Immersion Cooling Solution



HEADING FOR CHART/DIAGRAM

THE RISING SHARE OF U.S. POWER CONSUMED BY DATA CENTERS, 2023–2030



Conclusion: How it affects you?

Studies across the globe indicate 2-PIC can deliver:

- up to 40% lower Total Cost of Ownership (TCO) of hyperscale AI data centers compared to conventional cooling methods,
- up to 80% reduction of data center facilities-related operational energy costs,
- up to 17% lower total energy consumption,
- up to 16% reduction in capital expenditures.

Lowering energy costs means lowering electricity bills for communities living around data centers.

Driven by innovation and compelled by necessity, chipmakers and data center operators are preparing for the arrival of 1 MW IT racks around 2030. The Two-Phase Immersion Cooling technology is the most efficient cooling solution achieving the highest heat flux capabilities exceeding 300 W/cm².